Vaibhav Lalwani

Applied AI Engineer (Python, LLM systems)

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Student visa, moving to Graduate visa in early 2027 with full right to work. Eligible for the Skilled Worker new entrant rate.

SUMMARY

Applied AI engineer and MSc Advanced Data Science & AI student at the University of Liverpool, building production LLM applications, retrieval pipelines and agent products end to end, with evaluation and verification treated as part of the build. 2026 hackathon record includes the HydraDB x Connectors win (QueueProof) and the SharedOS Judges' Pick (Yuzu).

PROFESSIONAL EXPERIENCE

Full-Stack / AI Engineer | Meta Solution Technologies Apr - May 2026
Part-time, remote

- Built an AI-guided admissions workflow using Next.js, React, TypeScript and Python-backed REST APIs; integrated Supabase authentication and PostgreSQL/pgvector retrieval.
- Implemented document-processing and email workflows using Redis/BullMQ; handled validation, failure states and multilingual interfaces.

Software Engineer (Part-time) | Recruit Pilot Jan - Apr 2026
Part-time, remote

- Delivered React/TypeScript features and REST API integrations for a recruitment product, participating in code review and weekly release cycles.

Coding Instructor | RoboCode UK Aug 2026 - Present
Part-time, Liverpool, on-site

- Onboarded; compliance training complete; teacher induction in progress.

Front-End Developer Intern | Unified Mentor Mar - May 2025
Remote

- Built React features for a learner-facing platform, including lessons, progression views and REST-connected dashboards.

SELECTED ENGINEERING PROJECTS

HydraSentry | AI-agent memory security SaaS (FastAPI, Next.js) Source

Built a multi-tenant FastAPI backend with Supabase magic-link auth, per-user API keys, rate limiting and an MCP gateway that blocks poisoned-memory actions; 237 pytest tests pass, CI on every push.

Lacuna | Provenance-first memory for AI agents Source

Built temporal retrieval across web, CLI and MCP surfaces that abstains when evidence conflicts; 2,348 Vitest tests; includes a LongMemEval benchmark harness with recorded runs.

QueueProof | Cross-source evidence workspace Source

Retrieval pipeline with claim-level citations, degraded-source handling and an approval boundary for external writes; 654 Vitest tests pass; evaluation fixtures, grading scripts and measured results are published in the repo.

Kodro | Offline-first robotics learning studio (Python) Source

Pygame/Pymunk simulation app with Pydantic-validated models, an enforced 85% coverage gate and reproducible scenario evaluations committed as data.

SELECTED ACHIEVEMENTS

HydraDB x Connectors Winner (QueueProof) | SharedOS 2026 Judges' Pick (Yuzu) | HydraDB WikiThon Finalist (QyntraWiki Nexus) | Product Hunt #24 of the day (Kodro)

EDUCATION

MSc Advanced Data Science & Artificial Intelligence | University of Liverpool Expected Jan 2027

BCA | Christ University, Bengaluru 2022 - 2025

CERTIFICATIONS

Artificial Intelligence on Microsoft Azure (Microsoft, Coursera) - certificate ID VYN3SU1H9I3Z, Nov 2025

Microsoft Azure Machine Learning (Microsoft, Coursera) - certificate ID H4BG7UI694TN, Nov 2025

Natural Language Processing in Microsoft Azure (Microsoft, Coursera) - certificate ID 2W2FIN6EVBSL, Nov 2025

Computer Vision in Microsoft Azure (Microsoft, Coursera) - certificate ID OO1BFMY72ZDB, Nov 2025

Introduction to AI (Google, Coursera) - certificate ID ZDO3B89P0WKZ, Nov 2025

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

APIs and backend: FastAPI, REST, Node.js, PostgreSQL, Supabase, Redis, BullMQ, Docker

AI engineering: LLM APIs, RAG and retrieval evaluation, MCP servers, structured output validation with fallbacks, prompt/injection red-teaming

Experiment practice: benchmark fixtures, logged and reproducible runs, error-handling and timeout paths, cost-conscious local-model evaluation, Git, GitHub Actions, pytest, Vitest